

IT423 : Introduction to IT project management

2ND Semester 1447 H



Prepared by

Section No.	Students Names	Student ID
3583	Noura Alamro	444200941
	Shooq Alawadah	444201083
	Reema Alraqibah	444201069

Supervised by
Dr. Majdah Alshehri

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1. INTRODUCTION

The Kingdom of Saudi Arabia is home to many important historical and cultural destinations, such as AlUla and Historic Jeddah (Al-Balad), which reflect the Kingdom's rich heritage and attract visitors from both inside and outside the country. With the ongoing digital transformation aligned with Saudi Vision 2030, there is a strong focus on enhancing tourism. Technology can make visiting historical sites more organized, easier to manage, and more enjoyable. However, visitors often face challenges when planning their trips, as information is scattered across multiple platforms, making it difficult to find clear and reliable details in one place, which can make the planning process confusing and inefficient.

The IRTH Platform for Saudi Arabia is proposed as a unified mobile application focused on historical and cultural destinations across the Kingdom. The application will provide features such as a place planner, personalized recommendation system, issue reporting feature, and access to detailed destination information. The main goal of the application is to improve the visitor experience while supporting the promotion of Saudi heritage through practical and user-friendly digital solutions.

This report presents the project initiation, requirements, and scope, including the scope management plan, requirements management plan, main functional and non-functional requirements, scope statement, Work Breakdown Structure (WBS), and WBS dictionary.

2. PROJECT INITIATION

2.1. PROJECT DETAILED DESCRIPTION

- Problem

Saudi Arabia hosts several well-known historical destinations, including AlUla and Diriyah, attracting many visitors each year. However, visitors face multiple challenges due to the absence of a centralized system that provides complete and organized information. Currently, they must search across various sources to gather details about destinations, which is time-consuming, confusing, and sometimes unclear. Moreover, there is no single platform that enables visitors to plan trips, receive personalized recommendations, access important destination information, and report issues efficiently.

This lack of a structured system also limits communication between visitors and tourism authorities, making it harder to address problems that affect the visitor experience. These challenges highlight the need for a smart, unified platform that streamlines visit planning,

improves communication, and supports the promotion of Saudi heritage, providing a more convenient and satisfying experience for all visitors.

- Application solution

To address the identified challenges, the proposed solution is the development of **IRTH**, a **Smart Heritage Tourism Mobile Application** for Saudi Arabia. The platform is specifically designed to promote historical and cultural tourism by centralizing scattered information and providing a seamless user experience. While **IRTH** offers extensive data on iconic destinations like **AlUla and Diriyah**, it serves a strategic purpose in highlighting hidden heritage locations across the Kingdom, ensuring a more distributed and enriched tourism landscape.

The application is structured around four **Core System Modules** designed to fulfill the project requirements:

- **Trip Planner:** This module empowers users to manage their journeys by selecting specific visit dates and times, adding sites to a personal itinerary, and **booking visits** directly through the app. It ensures that the visitor's schedule is organized and efficient.
- **Recommendation System & Navigation:** To provide a personalized experience, this system suggests historical sites based on the **user's real-time location** and **individual interests**. To ensure seamless exploration, the application features an **interactive map** that allows users to view sites visually and receive **GPS directions** to their chosen locations. It also highlights points of interest during transit and provides detailed information relevant to each destination.
- **Issue Reporting:** This feature bridges the communication gap between tourists and authorities. Users can report immediate concerns such as **cleanliness issues, crowding, unavailable services, or outdated information**, ensuring high standards are maintained at heritage sites.
- **Admin Dashboard:** To complement the reporting feature, a dedicated interface is provided for administration. This allows officials to **receive, review, and act upon reports**, ensuring a continuous cycle of improvement and responsiveness.

By integrating these features, **IRTH** enhances the visitor journey while actively supporting the preservation and visibility of Saudi Arabia's cultural identity.

- Application Type

IRTH is designed as a **cross-platform mobile application** (compatible with both **iOS and Android**). This technical approach ensures maximum accessibility for a diverse user base, allowing tourists to access planning tools, navigation maps, and reporting features anytime and anywhere. The mobile-first strategy aligns with the need for a portable, smart tourism companion that assists the user throughout their physical journey.

2.2. PROJECT REQUIREMENTS

- Software tools
 - Microsoft Office (PowerPoint, Word, etc...)

- Dart Programming Language
- Visual Studio Code
- Project Plan 365
- Google Forms
- Lucidchart
- WhatsApp
- Flutter SDK
- GitHub
- Canva
- Figma
- Zoom
- Jira
- MySQL
- Hardware tools
 - Laptops for development and testing.
 - Smartphones (Android and iOS) for mobile application testing.
 - Server to host the application and the database.

2.3. PROJECT DATES AND DURATION

- Start Date: 1 January 2026
- End Date: 1 July 2026
- Duration: 6 Months

3. PROJECT SCOPE

3.1. SCOPE MANAGEMENT PLAN

The Scope Management Plan establishes the framework for defining, validating, and regulating the scope of the IRTH project throughout its execution. To ensure clarity from the outset, a comprehensive scope statement will be developed that captures the full vision of the mobile application. This document will articulate the project's purpose, key functionalities, and limitations in clear terms, covering aspects such as the trip planning system, recommendation engine, reporting mechanism, and information center. It will also define what is explicitly excluded from the project, such as developing a web version or handling financial transactions. By establishing this foundation, the scope statement provides a reference point that guides decision-making and keeps the team aligned with project objectives.

For structuring project work, a Work Breakdown Structure (WBS) will be developed using a top-down approach and progressive elaboration. The WBS will be expanded into major phases including research, design, development, testing, and deployment. Each phase will be further divided into specific tasks and subtasks until reaching manageable work packages that can be assigned and tracked. This hierarchical approach ensures complete coverage of all required activities while making complex work more understandable and manageable for team members.

Formal acceptance of deliverables will follow a structured validation process involving key stakeholders. Upon completion of major project components, the project supervisor and a select group of end-users will be invited to review the work against pre-established criteria documented in the scope statement. For the final application, this will involve hands-on testing of features across different mobile platforms to verify functionality and user experience. Successful completion of this review will result in formal sign-off documented with signatures and dates from the supervisor (on behalf of the project sponsor), to confirm official acceptance of the work delivered.

Scope change requests will be handled through a defined control mechanism to maintain project stability. When changes are proposed, they must be submitted in writing describing the nature of the change and its justification. The project team will assess each request considering factors such as impact on timeline, alignment with original goals, and resource availability. Based on this assessment, requests will either be approved and integrated into project plans or declined with clear reasoning communicated to the requester. This process ensures that only valuable and feasible changes are incorporated while protecting the project from uncontrolled expansion.

3.2. REQUIREMENTS MANAGEMENT PLAN

The Requirements Management Plan for the IRTH project defines the approach for handling requirements activities from identification through delivery. To plan requirements activities, the team will begin by analyzing the project description and then proceed with collecting requirements using methods such as interviewing stakeholders, holding focus groups, and conducting facilitated workshops. Through these collaborative discussions, we will

distinguish between functional requirements such as user registration, trip planning, and issue reporting, and non-functional requirements including response time, security, and usability. Each requirement will be prioritized using a simple classification system to distinguish between essential features that must be delivered and desirable enhancements that can be included if time permits. All identified requirements will be documented in a structured format that captures the description, priority level, and source of each item.

To track requirements activities throughout the project lifecycle, we will establish a Requirements Traceability Matrix using a shared spreadsheet. This matrix will serve as the central monitoring tool where each requirement is assigned a unique identifier, and its current status is regularly updated. Status categories will include proposed, approved, in progress, implemented, and verified. The matrix will also link each requirement to the corresponding work packages in the WBS and to the test cases that will verify successful implementation. During weekly team meetings, the project leader will review the matrix with all members to track progress, identify any requirements that are falling behind, and ensure that development efforts remain aligned with priorities.

To report requirements activities to stakeholders, progress summaries will be included in regular status reports prepared for the supervisor. These reports will provide a clear overview of requirements progress, including the number of requirements completed, those currently in development, and any pending items. The Requirements Traceability Matrix itself will be shared with the supervisor upon request, offering complete visibility into the status of each requirement and its journey through the project lifecycle. This reporting approach ensures transparent communication about requirements progress and supports informed decision-making throughout the project.

3.3. MAIN FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

FUNCTIONAL REQUIREMENTS:

1. The system shall allow users to create accounts and manage their cultural interests.
2. The user shall be able to create and customize a personalized itinerary for visiting heritage sites.
3. The system shall suggest heritage destinations based on the user's interests and previous visits.
4. The user shall be able to report any issues or maintenance needs at a site directly to the authorities.
5. The user shall be able to view comprehensive detailed destination information, including historical background, facilities, and services.
6. The user shall be able to search for historical and cultural sites by region, category, or name.
7. The user shall be able to view sites on an interactive map and get GPS directions to the locations.
8. The user shall be able to rate and review the sites visited to share their experience with others.

NON-FUNCTIONAL REQUIREMENTS :

1. Usability:
The interface must be user-friendly and support both Arabic and English languages.
2. Performance:
The application pages and search results must be loaded within less than 2 seconds.
3. Availability:
The platform shall be available 24/7 with a minimum up time of 99.5%.
4. Security:
All user data and communication with authorities must be encrypted and secured.

Requirements No.	Name	Category	Source	Status
FR1	User Account Management	Functional	Requirements Management Plan	Proposed - Registration and login module.
FR2	Trip Planning Tool	Functional	Project Detailed Description	Planned - Itinerary builder functionality.
FR3	Smart Recommendation	Functional	Project Detailed Description	Proposed - Personalization algorithm.
FR4	Issue Reporting	Functional	Project Detailed Description	Planned - Direct communication channel.
FR5	Information Hub	Functional	Project Detailed Description	Proposed - Content on historical sites.
FR6	Search & Filter	Functional	Application Solution	Planned - Search engine by categories.
FR7	Map & Navigation	Functional	Application Solution	Proposed - Interactive map integration.
FR8	User Feedback	Functional	Quality Standards	Planned - Rating and review system.
NFR1	Usability	Non-Functional	Requirements Management Plan	Planned - UI supporting Arabic/English.
NFR2	Performance	Non-Functional	Quality Standards	Planned - 2s response time target.
NFR3	Availability	Non-Functional	Quality Standards	Planned - Cloud-based uptime confirm.
NFR4	Data Security	Non-Functional	Quality Standards	Planned - Encryption for user profiles.

TABLE 1: REQUIREMENTS TRACEABILITY MATRIX (RTM) FOR IRTH

3.4. SCOPE STATEMENT

Scope Description	The IRTH platform is a mobile application developed to provide a centralized system for historical and cultural tourism in Saudi Arabia. The system enables users to access detailed information about heritage destinations, organize their visits through a trip planner, and receive personalized site recommendations. Additionally, it includes a reporting feature for direct communication with tourism authorities regarding site
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	issues. The project focuses on developing the core mobile application features and database integration. However, to maintain the project's focus within the given timeline, a web-based version and financial payment systems are currently outside the project scope.
Deliverables	The IRTH project will deliver a set of management and technical outputs throughout its duration. The final management deliverables consist of the Scope Management Plan, the Requirements Traceability Matrix (RTM), and the Work Breakdown Structure (WBS). Technically, the project will produce high-fidelity UI/UX designs created in Figma, followed by a fully functional mobile application built with Flutter and Dart. Additionally, a cloud-based database system using Firebase and MySQL will be delivered to store historical site data, along with a final demonstration of the working application.
Acceptance Criteria	The project will be successful when the application allows users to register, log in, and use the trip planner and recommendation features without errors. It must accurately display information and locations for popular heritage sites using integrated maps and GPS. The system must also demonstrate full bilingual support in Arabic and English and show a working reporting channel for site issues.
Constraints	The project is constrained by a strict 18-week timeline, requiring completion and submission by May 21, 2026. Technically, the development is limited to mobile platforms as defined in the project description. As a student project, the work must be completed using the team's available hardware and software resources without an external financial budget.
Assumptions	It is assumed that reliable and accurate historical data will be available from official Saudi heritage and tourism sources to feed the application. Another assumption is that the chosen development tools and cloud services will remain accessible and functional throughout the project duration. Finally, it is assumed that the team members have the necessary design and programming skills to complete the project tasks as planned.

TABLE 2:PROJECT SCOPE STATEMENT

3.5. WBS

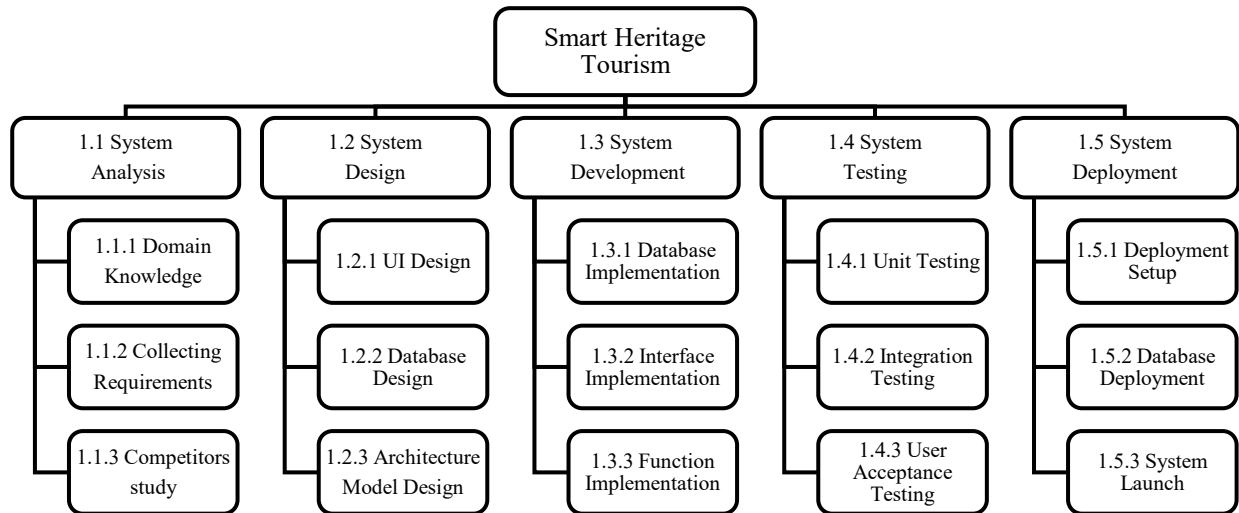


FIGURE 1: WBS CHART

3.6. WBS DICTIONARY

The following table illustrates the WBS Dictionary for selected tasks in the IRTH project.

WBS Dictionary
Project Title: IRTH
WBS Item Number: 1.1.2
WBS Item Name: Collecting Requirements
Description: The task of collecting requirements for the IRTH project involves conducting a structured analysis of tourists and tourism authorities to capture their needs and expectations. This process includes brainstorming sessions and reviewing the project description to identify key functional requirements such as trip planning, personalized recommendations, interactive maps, and issue reporting. Non-functional requirements including usability, security, performance, and availability will also be documented. The collected requirements will serve as the foundation for the system design phase. This task depends on the completion of Domain Knowledge analysis (WBS 1.1.1) and must be completed before system design tasks (WBS 1.2).

TABLE 3:COLLECTING REQUIREMENTS DICTIONARY

WBS Dictionary
Project Title: IRTH
WBS Item Number: 1.2.1
WBS Item Name: UI Design
Description: The task of UI design in the IRTH project focuses on developing structured, user-friendly, and visually clear interfaces that enhance the overall visitor experience. The design process ensures bilingual support (Arabic/English) and intuitive navigation. Interfaces will include registration and login screens, trip planner management, recommendation display, interactive maps with GPS navigation, and issue reporting features. High-fidelity prototypes created using Figma will guide the development phase. This task depends on the completion of the Collecting Requirements task (WBS 1.1.2) and feeds directly into Interface Implementation (WBS 1.3.2).

TABLE 4 UI DESIGN DICTIONARY

WBS Dictionary
Project Title: IRTH
WBS Item Number: 1.4.1
WBS Item Name: Unit Testing
Description: The task of unit testing in the IRTH project involves verifying individual modules of the mobile application before system integration. Core components including user account management, trip planner operations, recommendation logic, interactive map navigation, and issue reporting will be tested independently to ensure correct functionality. Predefined test cases will be used to detect defects early and confirm compliance with defined requirements. This task depends on the completion of Function Implementation (WBS 1.3.3) and must be completed before Integration Testing (WBS 1.4.2).

TABLE 5 UNIT TESTING DICTIONARY

4. PROJECT PLANNING (WBDS TABLE)

Phase Name	Start Date	End Date	Duration	Qualifications	Tools	Deliverables
System Analysis	1-Jan-2026	28-Jan-2026	20 days	-Business Analyst -Data Analyst	-Microsoft Word -Google Forms -Microsoft Excel -Jira -Project Plan 365 -Zoom	-Stakeholder Analysis -Functional Requirements -Nonfunctional Requirements -Project Budget
System Design	29-Jan-2026	25-Feb-2026	20 days	-UI Designer -System Architect -Database Designer	-Canva -Lucidchart -Figma -Zoom -Microsoft word	-Interface Design -system Architecture -Database Design
System Development	26-Feb-2026	29-May-2026	67 days	-System Developers (Flutter, Backend & Database)	-Dart Programming Language -Flutter SDK -Visual Studio Code -GitHub -Jira -MySQL	-Mobile Application -User Registration & Login Module -Trip Planner Module - Recommendation Engine -Map & GPS Integration -Issue Reporting Module -Database Implementation

System Testing	1-Jun-2026	26-Jun-2026	20 days	-Developer -Analyst	-Flutter Testing Tools -Jira (Bug Tracking) -Smartphones -MySQL	-Unit Testing Report -Integration Testing Report -User Acceptance Testing (UAT) -Final Validated Application
System Deployment	27-Jun-2026	1-Jul-2026	4 days	-Development Team -Project Leader	-Server Hosting Platform -GitHub -Flutter SDK -MySQL -Zoom	-Deployed Application -Final Project Report -Final Database Setup -Final Presentation -Demo Application

TABLE 6: WBDS TABLE

5. TECHNICAL OUTCOMES AND CHARTS

5.1. NETWORK DIAGRAM WITH CRITICAL PATH

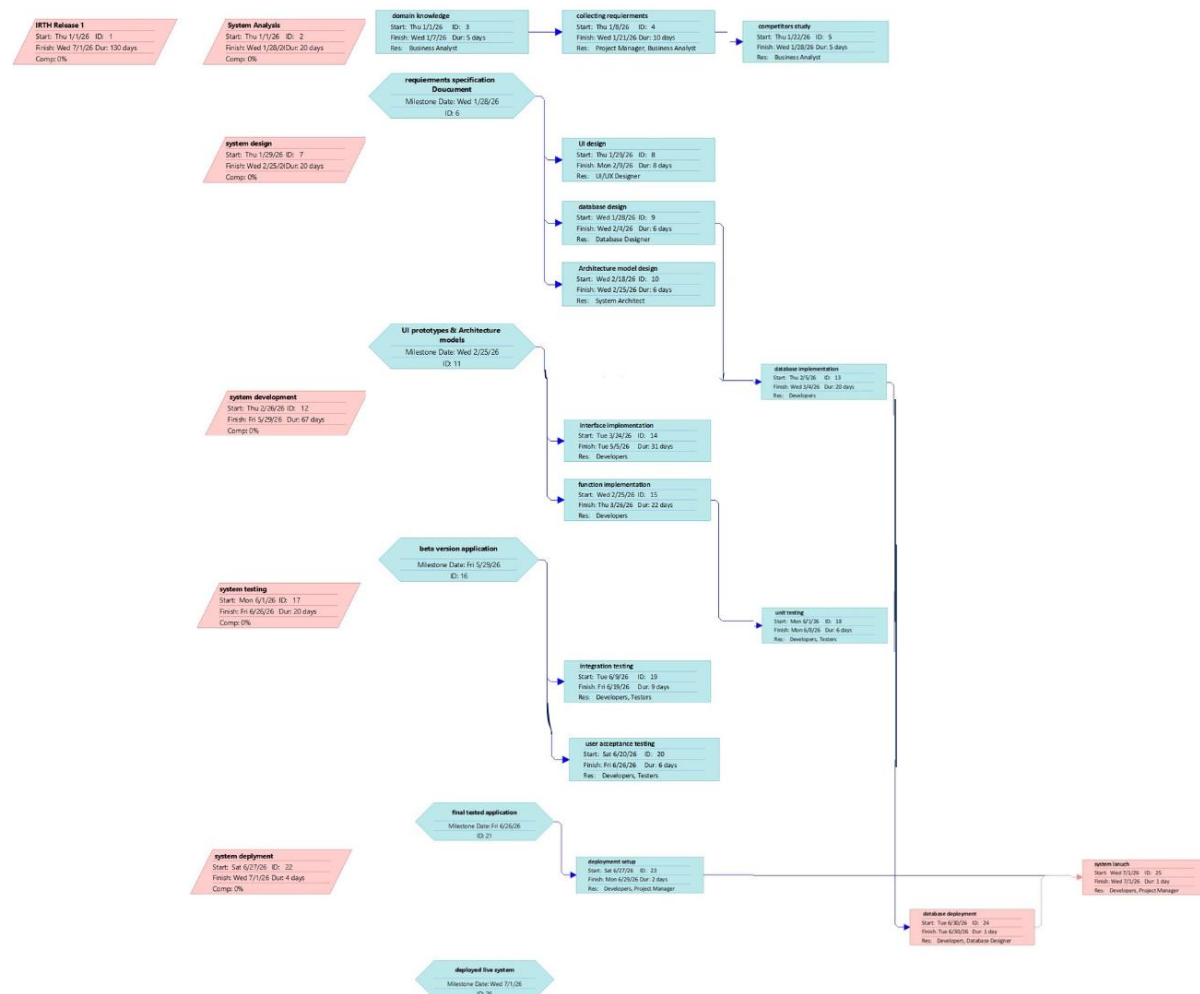


FIGURE 2 NETWORK DIAGRAM

critical path:

The project's critical path consists of the System Analysis, System Design, System Development, System Testing, and System Deployment phases. These activities follow a strict finish-to-start sequence with no overlapping tasks. As a result, they collectively determine the project's total duration of 130 days, and any delay in any of these phases will directly delay the overall project completion date.

5.2. A COMPLETE GANTT CHART

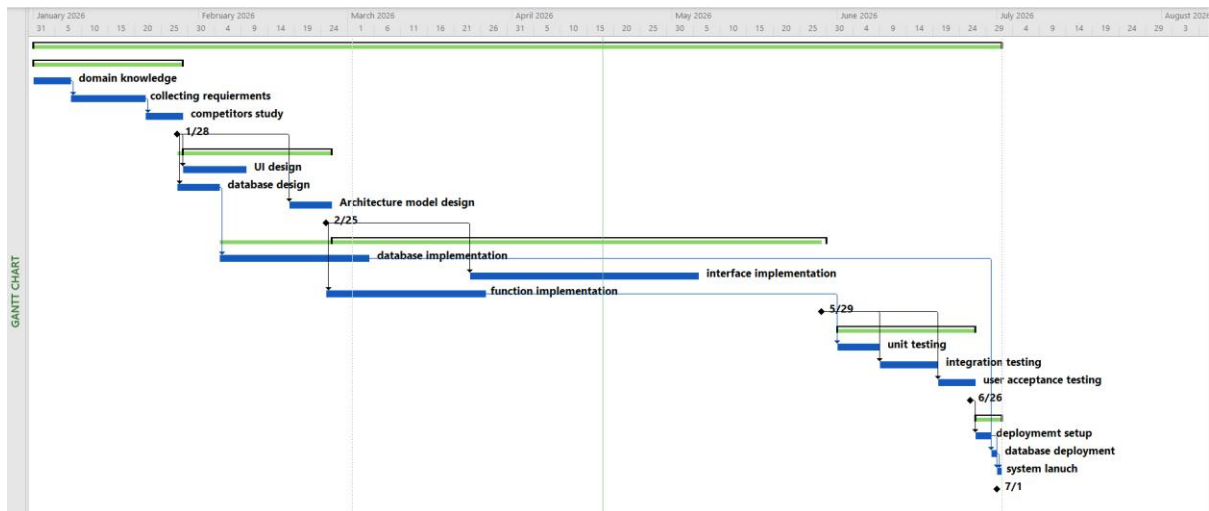


FIGURE 3:GANTT CHART

5.3. PROJECT TIMELINE

	Task Mode	Task Name	Duration	Start	Finish	Predecessor	Resource Names
1		IRTH Release 1	130 days	Thu 1/1/26	Wed 7/1/26		
2		System Analysis	20 days	Thu 1/1/26	Wed 1/28/26		
3		domain knowledge	5 days	Thu 1/1/26	Wed 1/7/26		Business Analyst
4		collecting requirements	10 days	Thu 1/8/26	Wed 1/21/26	3	Project Manager,Business Analyst
5		competitors study	5 days	Thu 1/22/26	Wed 1/28/26	4	Business Analyst
6		requirements specification Document	0 days	Wed 1/28/26	Wed 1/28/26		
7		system design	20 days	Thu 1/29/26	Wed 2/25/26		
8		UI design	8 days	Thu 1/29/26	Mon 2/9/26	6	UI/UX Designer
9		database design	6 days	Wed 1/28/26	Wed 2/4/26	6	Database Designer
10		Architecture model design	6 days	Wed 2/18/26	Wed 2/25/26	6	System Architect
11		UI prototypes & Architecture models	0 days	Wed 2/25/26	Wed 2/25/26		
12		system development	67 days	Thu 2/26/26	Fri 5/29/26		
13		database implementation	20 days	Thu 2/5/26	Wed 3/4/26	9	Developers
14		interface implementation	31 days	Tue 3/24/26	Tue 5/5/26	11	Developers
15		function implementation	22 days	Wed 2/25/26	Thu 3/26/26	11	Developers
16		beta version application	0 days	Fri 5/29/26	Fri 5/29/26		
17		system testing	20 days	Mon 6/1/26	Fri 6/26/26		
18		unit testing	6 days	Mon 6/1/26	Mon 6/8/26	15	Developers,Testers
19		integration testing	9 days	Tue 6/9/26	Fri 6/19/26	16	Developers,Testers
20		user acceptance testing	6 days	Sat 6/20/26	Fri 6/26/26	16	Developers,Testers
21		final tested application	0 days	Fri 6/26/26	Fri 6/26/26		
22		system deployment	4 days	Sat 6/27/26	Wed 7/1/26		
23		deployment setup	2 days	Sat 6/27/26	Mon 6/29/26	21	Developers,Project Manager
24		database deployment	1 day	Tue 6/30/26	Tue 6/30/26	13	Developers,Database Designer
25		system launch	1 day	Wed 7/1/26	Wed 7/1/26	23,24	Developers,Project Manager
26		deployed live system	0 days	Wed 7/1/26	Wed 7/1/26		

FIGURE 4:PROJECT TIMELINE

5.4. TASK ALLOCATION SCHEDULE

		Task Mode	Task Name	Duration	Start	Finish	Predecessor	Resource Names
1			IRTH Release 1	130 days	Thu 1/1/26	Wed 7/1/26		
2			System Analysis	20 days	Thu 1/1/26	Wed 1/28/26		
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4			collecting requierments	10 days	Thu 1/8/26	Wed 1/21/26	3	Project Manager,Business Analyst
5			competitors study	5 days	Thu 1/22/26	Wed 1/28/26	4	Business Analyst
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7			system design	20 days	Thu 1/29/26	Wed 2/25/26		
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24			database deployment	1 day	Tue 6/30/26	Tue 6/30/26	13	Developers,Database Designer
25			system lanuch	1 day	Wed 7/1/26	Wed 7/1/26	23,24	Developers,Project Manager
26			deployed live system	0 days	Wed 7/1/26	Wed 7/1/26		

FIGURE 5:TASK ALLOCATION SCHEDULE

5.5. TASK DEPENDENCIES TABLE

		Task	Task Name	Duration	Start	Finish	Predecessor	Resource Names
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26			deployed live system	0 days	Wed 7/1/26	Wed 7/1/26		

FIGURE 6:TASK DEPENDENCIES TABLE

6. COST ESTIMATION

6.1. DEVELOPMENT COST

- AFP (Adjusted Function Points)

To calculate the Adjusted Function Points (AFP), the following formulas are used:

- Total Function Points (FP) = Inputs + Outputs + Files + Inquiries + Interfaces = $22 + 18 + 20 + 14 + 12 = \mathbf{86 \text{ FP}}$
- Adjustment Factor (AF = **1.15**) is selected to reflect moderate system complexity, including external interfaces (GPS services), data processing (recommendation system), and user interaction features.
- Adjusted Function Points (AFP) = $86 \times 1.15 = \mathbf{99 \text{ FP}}$

- **ASLOC (number of lines of generated code)**

Since the project is developed using Dart (Flutter), the average is:

$1 \text{ FP} \approx 50 \text{ LOC}$

$\text{ASLOC} = 99 \times 50 = \mathbf{4,950 \text{ LOC}}$

- **Number of staff – Month**

Productivity = **450 LOC / Staff-Month**

Staff-Months = $4,950 / 450 = \mathbf{11 \text{ Staff-Months}}$

Estimated team size = $11 / 6 = 1.833 \approx 2 \text{ Employees}$

- **Salary for development staff**

Salary for development staff = (Number of Employees $\times$ salary average $\times$ project duration in months) = $2 \times 5,000 \times 6 = \mathbf{60,000 \text{ SAR}}$

Where 5,000 SAR per staff-month.

- **Effort estimation using COCOMO model**

We are using the Semi-Detached COCOMO model to estimate the effort required for developing our project. This model was selected after analyzing the nature of our system, the team size, and the complexity level of the project. The Semi-Detached model is the most suitable due to the following reasons:

1. Project Size and Complexity:

Our system is considered medium-sized with moderately complex features such as trip planning, personalized recommendations, interactive maps with GPS, and issue reporting.

The integration of external services like maps and location tracking adds additional complexity. This aligns with the Semi-Detached model, which is designed for projects with intermediate complexity.

2. Team Experience:

The development team consists of members with mixed experience levels in mobile application development using Flutter and Dart. Some team members have prior experience, while others are still learning. This matches the Semi-Detached model assumptions, which consider teams with varying levels of experience.

3. Project Type and Constraints:

IRTH is a mobile application that includes both frontend and backend components, along with database integration and external APIs. It is more complex than simple Organic projects, but it does not involve strict hardware or real-time constraints like Embedded systems. Therefore, the Semi-Detached model is the most appropriate.

4. Well-Defined Requirements and Documentation:

The system requirements and main functionalities are clearly defined, including user roles, trip planning, recommendation system, and reporting features. The Semi-Detached model fits projects with stable and moderately complex requirements.

• Select the constant values

Based on the Semi-Detached model:

- $a = 3.0$
- $b = 1.12$

$$\text{KLOC} = \text{ASLOC} / 1000 = 4,950 / 1000 = \mathbf{4.95 \text{ KLOC}}$$

• Cost drivers

Cost Driver	Rating	Effort Multiplier
RELY (Required Software Reliability)	Low	0.88
DATA (Database Size)	Nominal	1.00
CPLX (Product Complexity)	High	1.15
TIME (Execution Time Constraint)	Nominal	1.00
STOR (Main Storage Constraint)	Nominal	1.00
ACAP (Analyst Capability)	High	0.86
PCAP (Programmer Capability)	Nominal	1.00

AEXP (Applications Experience)	Nominal	1.00
TOOL (Use of Software Tools)	High	0.91
SCED (Required Development Schedule)	Nominal	1.00

TABLE 7 COST DRIVERS

Note: “RELY is considered low since the system is not safely critical.”

Effort Adjustment Factor (EAF) = Product of all effort multipliers:

$$\text{EAF} = 0.88 \times 1.00 \times 1.15 \times 1.00 \times 1.00 \times 0.86 \times 1.00 \times 1.00 \times 0.91 \times 1.00$$

Step by step calculation:

1. $0.88 \times 1.15 = 1.012$
2. $1.012 \times 0.86 = 0.870$
3. $0.870 \times 0.91 = 0.79$

So:

$$\text{EAF} = 0.79$$

• Effort using COCOMO model

The effort is calculated using the following formula:

$$\text{Adjusted Effort} = a \times (\text{KLOC})^b \times \text{EAF}$$

By substituting the values:

$$\begin{aligned} \text{Adjusted Effort} &= 3.0 \times (4.95)^{1.12} \times 0.79 \\ \text{Adjusted Effort} &= 3.0 \times 6.00 \times 0.79 \end{aligned}$$

$$\text{Adjusted Effort} = \mathbf{14.22 \text{ Staff-Months}}$$

COCOMO Cost:

$$\text{COCOMO Cost} = 14.22 \times 5,000 \text{ SAR} = 71,100 \text{ SAR}$$

Estimation Method	Staff-Months	Cost (SAR)
Bottom-Up (Function Point)	11	60,000
COCOMO (Semi-Detached)	14.22	71,100

TABLE 8 ESTIMATION METHOD

Difference Calculation:

$$((71,100 - 60,000) / 71,100) * 100 = 15.61\%$$

Since the difference is less than 20%, both estimates are consistent, and we will use the higher estimate:

Development Cost = 71,100 SAR

Basis of Estimates

The following assumptions were used in developing the cost estimates for IRTH :

- **Productivity rate:** 450 LOC per staff-month is assumed based on the team's intermediate experience with Flutter and Dart, and the availability of reusable Flutter widgets.
- **Salary rate:** 5,000 SAR per staff-month is assumed based on local entry-level rates for mobile developers reflecting that actual development is performed by students, not industry rates. This covers development effort only, excluding benefits or overhead.
- **Function point to LOC conversion:** 50 LOC per FP is used, which is standard for Java and Dart (Flutter) as referenced in the QSM Function Point Languages Table.
- **Adjustment factor (AF = 1.15):** Reflects moderate system complexity due to GPS integration, a personalized recommendation algorithm, and bilingual (Arabic/English) UI support.

6.2. TOOLS COST

Type	Tool / Platform	Cost (SAR)
Software	Microsoft Office (PowerPoint, Word, Excel)	85 × 6 months = 510
	Project Plan 365	1,000
	Visual Studio Code	Free
	Dart Programming Language	Free
	Flutter SDK	Free
	Google Forms	Free

	Lucidchart	Free
	WhatsApp	Free
	GitHub	Free
	Canva	Free
	Figma	Free
	Zoom	Free
	Jira	Free
	MySQL	Free
Hardware	HP Pavilion Laptops for development and testing	3,500 * 4 = 14,000
	Smartphone (Samsung Galaxy A51 or similar)	1,500 * 2 = 3,000
	Server to host the application and the database	1,500
Total Tools Cost		20,010 SAR

TABLE 9 TOOLS COST

Hardware cost is estimated assuming that the required devices are not already available.
Otherwise, these costs can be excluded as organizational resources

6.3. TOTAL COST

The total project cost is calculated by adding the development cost and the tools cost:

Total Cost = Development Cost + Tools Cost = 71,100 + 20,010 = **91,110 SAR**

This represents the overall estimated budget required to complete the IRTH project.

7. QUALITY ATTRIBUTES

Quality Attribute	Description	Measurement	Assessment
Usability	The system shall provide a simple, intuitive, and user-friendly interface with full support for both Arabic and English languages to enhance user experience.	Users can complete main tasks within 10 minutes	Conduct User Acceptance Testing (UAT) with a sample group of end-users. Use a Checksheet to record and analyze the exact time taken to complete

			core tasks, ensuring the 10-minute threshold is met.
Performance	The system shall respond quickly to user actions, with most operations such as page loading and search results completed within 2 seconds.	Response time $\leq$ 2 seconds	Perform performance testing using Flutter DevTools under normal load (50 concurrent users) and peak load (100 concurrent users). Measure API response times and screen rendering.
Availability	The platform shall be available 24/7 with a minimum uptime of 99.5% of the total operational time, ensuring uninterrupted access to application services.	System uptime $\geq$ 99.5%	Monitor uptime using Firebase hosting logs or server monitoring tools (e.g., UptimeRobot) for 30 consecutive days during the testing phase. Availability = $(\text{Up Time} / \text{Total Time}) \times 100$.
Security	The system must protect user data by applying strong encryption for passwords and securing communications between the app and servers.	Number of detected vulnerabilities categorized by severity (Low, Medium, High, Critical)	No High or Critical vulnerabilities are allowed, and all identified issues must be resolved before deployment.

TABLE 10 QUALITY ATTRIBUTES

REFERENCES

- [1] QSM, "Function Point Languages Table," QSM, Inc. [Online]. Available: <https://www.qsm.com/resources/function-point-languages-table>. [Accessed: 20-Apr-2026].
- [2] B. W. Boehm, *Software Engineering Economics*. Englewood Cliffs, NJ: Prentice-Hall, 1981.